A MINI PROJECT REPORT

ON

FACE RECOGNITION SYSTEM

Submitted in partial fulfilment of the requirement for the award of Degree

BACHELOR OF THECHNOLOGY

IN

COMPUTER SCINECE AND TECHNOLOGY(AIML)

BY

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2024-2025

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CERTIFICATE

This is to certify that the project report entitled **“FACE RECOGNITION SYSTEM” R.POOJITHA (21N71A6632), A.GEETHIKA(21N71A6601), B.PRANAY KUMAR(21N71A6655), J.NAVEEN(21N71A6613)** or the partial fulfilment of the requirement for the award of B.Tech. Degree in **COMPUTER SCIENCE AND ENGINEERING(AI & ML),** JNTUH University Hyderabad, for the academic year 2024-2025

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EXTERNAL EXAMINER

DECLARATION

We here declare that the project report entitled **"FACE RECOGNITION SYSTEM"** submitted to the Department of **COMPUTER SCIENCE AND ENGINEERING (AI & ML)** in partial fulfilment of requirements for the award of the degree of **BACHELOR OF TECHNOLOGY.** This project is the result of our own effort and that it has not been submitted to any other University or Institution for the award of any degree or diploma other than specified above.

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ABSTRACT

In today's digital age, optimizing facial emotions of our own face. The Facial Recognition-Based on Facial expressions, developed using Machine Learning, represents a cutting-edge solution designed to recognize facial expressions. This innovative application leverages state-of-the-art facial recognition technology, backed by machine learning algorithms, to recognise facial emotions like sadness, happiness, angriness etc.. quickly and accurately. By capturing and analysing unique facial features, the system ensures a secure and efficient. The system's primary features include real-time facial expressions, automatic data integration with the persons database, and robust security protocols to safeguard sensitive information. Additionally, it provides instructors with comprehensive, empowering them to make data-driven decisions to enhance facial expressions.

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INTRODUCTION

1.1 Introduction

Facial recognition technology relies on the intricate patterns of a person's facial characteristics, such as the arrangement of eyes, nose, mouth, and other distinguishing features. These systems capture and process facial data using sophisticated algorithms, enabling them to perform tasks like identifying individuals in a crowd, verifying identities for secure access, or even assisting in law enforcement efforts.

The possibilities presented by facial recognition are substantial, but they also come with significant responsibilities and considerations. Striking a balance between the convenience and utility of facial recognition and safeguarding individual privacy and civil liberties is an ongoing challenge. Ethical and legal issues surrounding data privacy, bias mitigation, and transparency have sparked intense debates and have led to the development of guidelines and regulations to govern the use of this technology.

1.2 Purpose

Facial recognition systems designed to detect and interpret facial expressions serve a multifaceted array of purposes across diverse domains. These systems primarily excel in understanding and decoding human emotions, offering applications ranging from healthcare to security, education, retail, and entertainment. By accurately identifying emotions like happiness, sadness, anger, and surprise, these systems facilitate enhanced human-computer interaction, aid in healthcare diagnostics, improve security measures by detecting distress or suspicious behaviour, gauge student engagement in educational settings, provide valuable insights for retail and marketing, and enhance gaming experiences. Additionally, they hold potential in law enforcement and forensics by aiding in suspect identification or assessing veracity through micro-expression analysis. However, ethical considerations, including privacy, consent, and potential misuse, remain critical aspects to navigate when implementing such systems due to their sensitive nature in analysing personal emotional data.

1.3 Problem Statement

Privacy: Facial recognition systems often involve the collection and analysis of biometric data, raising serious privacy concerns. Individuals may not always be aware of when and how their facial data is being collected and used, necessitating robust privacy protections.

Security: While facial recognition can enhance security measures, it also introduces vulnerabilities, as the technology can be exploited for unauthorized access or surveillance. Protecting against misuse and data breaches is a significant challenge.

Bias Mitigation: Facial recognition algorithms have been found to exhibit biases, particularly against certain racial and gender groups. Addressing these biases is essential to prevent discrimination and ensure fairness.

Transparency: The lack of transparency in how facial recognition systems work, the data they use, and the decisions they make can erode trust. Transparent and explainable algorithms are needed to build public confidence.

CHAPTER –2

SYSTEM SPECIFICATION

2.1 HARDWARE REQUIREMENTS

- ❖ **System** : Intel i5
- ❖ **Hard Disk** : 128GB.
- ❖ **Monitor** : 15 LED.
- ❖ **Mouse** : Optical Mouse.
- ❖ **RAM** : 8GB or More.

2.2 SOFTWARE REQUIREMENTS

- ❖ **Operating system** : Windows 10.
- ❖ **Coding Language** : Python.
- ❖ **Facial Detection** : OpenCV , Dlib

CHAPTER - 3

SOFTWARE AND HARDWARE SPECIFICATIONS

3.1 REQUIREMENT ANALYSIS

The project involved analysing the design of few applications so as to make the application more users friendly. To do so, it was really important to keep the navigations from one screen to the other well-ordered and at the same time reducing the amount of typing the user needs to do. In order to make the application more accessible, the browser version had to be chosen so that it is compatible with most of the Browsers.

3.2 REQUIREMENT SPECIFICATION

3.2.1 Functional Requirements

- Graphical User interface with the User.

3.2.2 Software Requirements

For developing the application, the following are the Software Requirements:

1. Python

Operating Systems Supported

1. Windows 10 64 bit OS

Technologies and Languages used to Develop

1. Python

Debugger and Emulator

- Any Browser (Particularly Chrome)

3.2.3 Hardware Requirements

For developing the application, the following are the Hardware Requirements:

- Processor: Intel i5
- RAM
- Space on Hard Disk: 128GB

CHAPTER - 4

LITERATURE SURVEY

1)Face Detection and recognition using viola-jones algorithm and fusion of PCA and ANN AUTHORS: Deshpande, N.T & Ravishankar, S

The combination of the Viola-Jones algorithm, Principal Component Analysis (PCA), and Artificial Neural Networks (ANNs)provides an effective approach for face detection and recognition. The Viola-Jones algorithm is used for detecting faces in real-time by utilizing Haar-like features and an efficient cascade classifier. Once faces are detected, PCA reduces the dimensionality of face images, extracting significant features as eigenfaces, which helps speed up the recognition process. The extracted features are then classified using ANNs, particularly Multilayer Perceptron (MLP)networks, which are trained to map these features to specific identities. By combining PCA for dimensionality reduction and ANNs for complex classification, this fusion enhances both the accuracy and computational efficiency of the system. The approach is useful in applications such as surveillance, biometric identification, and human-computer interaction. Future advancements could explore combining face recognition with other biometric modalities, such as voice or fingerprint recognition.

2)PCA algorithm for Human Face Recognition. Ohol,M.R.M,& Ohol,M.S.R.

AUTHORS: Ohol, M.R.M,&Ohol, M.S.R

Principal Component Analysis (PCA) is a statistical technique used for dimensionality reduction and feature extraction in face recognition systems. It identifies the directions (principal components) along which the variance in the dataset is maximized. In the context of face recognition, PCA transforms high-dimensional face image data into a lower-dimensional subspace while retaining the most significant features. Each face image is represented as a vector and normalized. PCA computes the covariance matrix of the dataset and its eigenvalues and eigenvectors. The eigenvectors corresponding to the largest eigenvalues are selected to form the reduced feature space. These eigenvectors, often referred to as "eigenfaces," capture the dominant facial features. During recognition, a new face image is projected onto the PCA.

3) Digital image Processing

AUTHORS : R.C Gonzalez and R.E

Image processing refers to the manipulation and analysis of digital images using various algorithms and techniques to enhance, transform, or extract information from them. It involves a series of steps, which can range from simple enhancements to complex analyses. main categories and techniques commonly used in digital image processing:

1. **Image Enhancement**:

Contrast Adjustment: Enhances the difference between light and dark areas in the image.

Brightness Adjustment: Modifies the overall lightness or darkness of the image.

Noise Reduction: Filters out unwanted noise (random variations) in an image, often using techniques like Gaussian blur or median filters.

Sharpness Enhancement: Improves the detail and edges in an image by enhancing the high-frequency components.


2. Image Transformation:

Geometric Transformations: Scaling,rotation,and Translation of images to adjust their shape or position

Original Image After rotation of 45°

3.Image Compression:

Lossy Compression: Techniques like JPEG that reduce file size by sacrificing some image quality, usually imperceptible to the human eye.

Lossless Compression: Techniques like PNG that reduce file size without losing any image data.

Wavelet Compression: Used for multi-resolution analysis, compressing images in a way that allows for progressive refinement.

REAL IMAGE

COMPRESSED IMAGE

4.Morphological Operations

Dilation and Erosion: image processing operations that change the size, shape, and connectivity of objects in an image.

Opening and Closing: Processes that help remove noise or small objects from binary images.

Original image

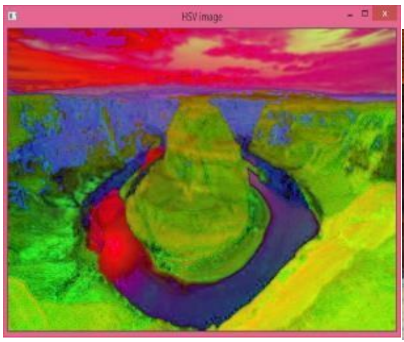

Dilated image


Eroded image


Opened image


Closed image


Colour Image Processing: Colour Space Transformation: Converting an image from one colour space to another, such as RGB to HSV, for easier manipulation (e.g., thresholding in hue or saturation channels).



Colour Filtering: Isolating specific colours in an image, often used in object detection or tracking.

4. Image segmentation through unified combinational user input.

AUTHORS : Yang,W.Cai,J.Zheng,j.Luo,J.

Image segmentation is a crucial task in computer vision, which involvespartitioning an image into distinct regions or segments to simplify its analysis. Traditional methods, such as thresholding or clustering, can be effective, but they often struggle with complex, real-world images where objects may overlap or have unclear boundaries. A more sophisticated approach to image segmentation leverages unified combinatorial user inputs, enabling better control and precision in segmenting an image. Unified combinatorial user inputs involve a system where the user provides a set of combined inputs that guide the segmentation process. These inputs can be a combination of several types, such as object boundary marking, region-of interest selection, or even user- driven constraints that define the relationships between different segments. By integrating multiple forms of user interaction, the system can learn more complex patterns and apply these learned patterns to segment an image more effectively. One of the main advantages of this approach is its ability to incorporate human expertise directly into the process, allowing the system to handle more intricate details and edge cases that purely automated methods may miss. a user might draw rough boundaries around objects in the image, and the system could use these inputs to refine the segmentation, correcting errors or filling in gaps that would be difficult for a fully automated algorithm. Furthermore, the combinatorial aspect of these inputs allows the system to handle diverse image types and complexities. A single input method might not work well across all images, but by combining different inputs, the system becomes more versatile and adaptive to the varying characteristics of each image.

A colour image of a cat sitting on a couch

Segmentation Task:

Separate the cat from the couch and the background Three separate images:

1. Cat: The cat is extracted from the original image, with a clear boundary around its fur, eyes, and whiskers.

2. Couch: The couch is extracted, including its texture, pattern, and shape.

3. Background: The background is extracted, including the wall, floor, and any other objects.

Segmentation Techniques Used:

Edge Detection: To identify the boundaries between the cat, couch, and background.

Thresholding: To separate the cat from the couch and background based on colour and texture.

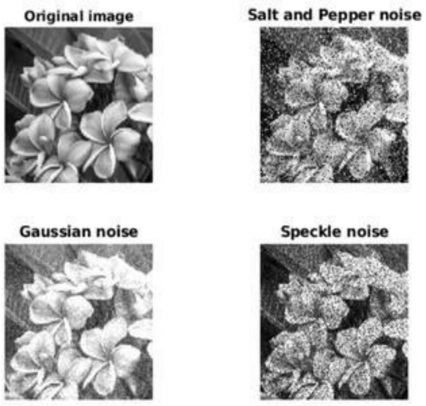
Clustering: To group similar pixels together and form regions.

Contour Detection: To identify the shape and boundary of the cat, couch, and background.

Image segmentation step by step process :

Image Preprocessing: Image made up of pixels, which are tiny boxes representing the colour and brightness values at that point in the image. Image processing involves handling these pixels in a desired manner to achieve what is required for the image. Most of the common operations performed on a digital image include filtering, enhancement, restoration, etc.

Filtering is a process of eliminating unwanted noise from an image. It is done by applying a filter that adjusts the image's pixel values. Based on the type of filter, they can be used for a wide range of applications. They can be designed to remove specific types of noise, such as Gaussian noise, salt-and-pepper noise, or speckle noise. The filters that help in removing the above-mentioned noises include the median filter, the mean filter, and the Gaussian filter.



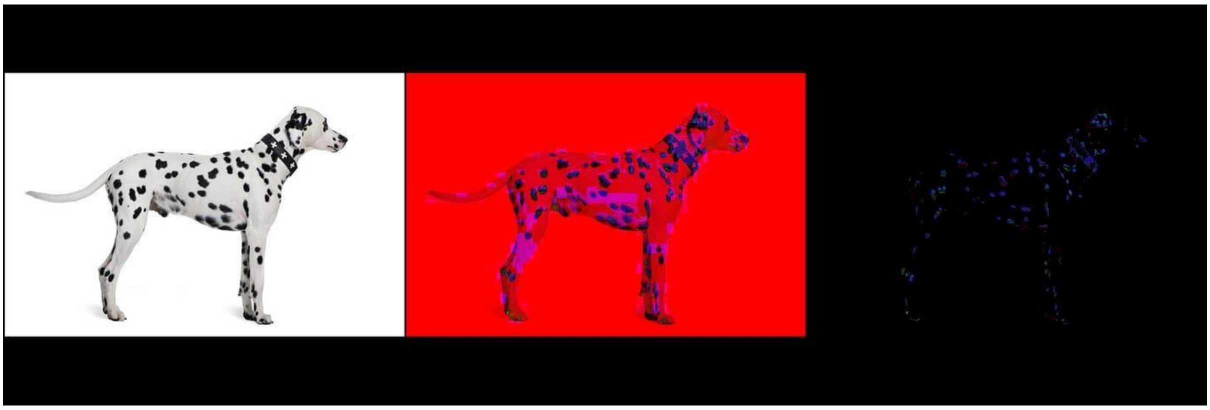
Enhancement is one process that can improve the quality of an image. It is done by modifying the brightness or contrast of the image. These techniques may be simple, like adjusting the brightness and contrast using a histogram, or more complex, like using algorithms to enhance the edges and textures in an image.

Noice reduction:

Noice reduction is the process of recovering an image that some noise or other artifacts may degrade. The techniques involve using mathematical methods to estimate the original image from the corrupted version. It is done using techniques such as deconvolution, which is used to get the original image from a blurred version, or denoising, which is used to remove noise from an image.

Image feature extraction:

Colour Features: Histogram of colours, colour moments, and histogram. **Texture Features**: GLCM, LBP, Gabor filters, and statistical texture features. **Shape Features**: Edge detection

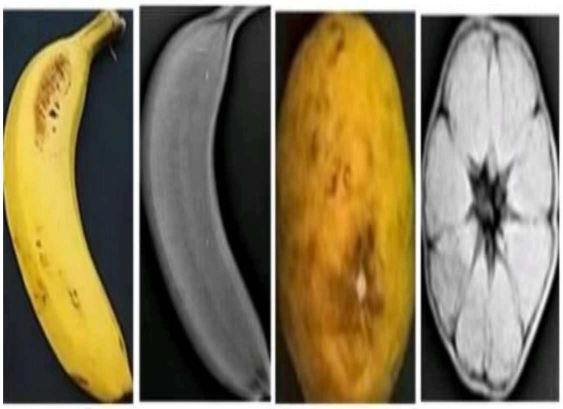


methods like texture, colour, shape, and keypoint-based feature

Feature selection:

feature selection for an image classification task, where we want to classify images of fruits (e.g., oranges, bananas) based on various extracted features like colour, texture, and shape. Image Dataset and Feature Extraction

Two different types of fruits: oranges and bananas. We want to classify these images based on their features. features from each image:



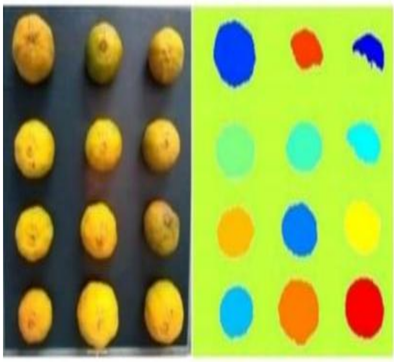


Image classification :

Image classification is a task in computer vision where an algorithm is used to categorize an image into a predefined class or label based on its content. The goal is to assign a specific label to an image based on the objects, scenes, or patterns present in it.

This process involves training a machine learning or deep learning model, typically using large datasets of labelled images. Popular techniques used for image classification include Convolutional Neural Networks (CNNs), which are particularly effective for tasks involving image recognition due to their ability to capture spatial hierarchies and features in images.

we classify images of three animals cats and dogs. The goal is to train a model that can correctly classify images into one of these categories.

Step-by-Step Process:

Data Collection:

Images: You gather a dataset of images of cats and dogs. The dataset should include various pictures of these animals in different poses, lighting conditions, and backgrounds.

Labels: Each image in the dataset is labelled as either “Cat,” “Dog,” or “Horse.”

Data Preprocessing:

Resizing: All images are resized to a consistent size , which is required for most machine learning models.

Normalization: Pixel values of the images are normalized to a range between 0 and 1 by dividing each pixel.

Augmentation: You apply random transformations (e.g., rotation, flipping, zooming) to the images in the training set. This helps the model generalize better to new data.

Choosing a Model:

Convolutional Neural Network (CNN) is chosen because it is highly effective for image classification tasks.

The model could consist of:

Convolutional layers: Detect edges, shapes, and textures.

Pooling layers: Reduce image dimensions while retaining important features.

Softmax output layer: Used to predict the class probabilities for "Cat," "Dog," or Model **Training:**

The model is trained using the training images and their labels (e.g., "Cat” and "Dog”).

During training, the model learns to recognize the distinguishing features of each animal, such as:

Cats: Pointed ears, smaller size, and whiskers.

Dogs: Different shapes and sizes, but often with rounded ears and snouts.

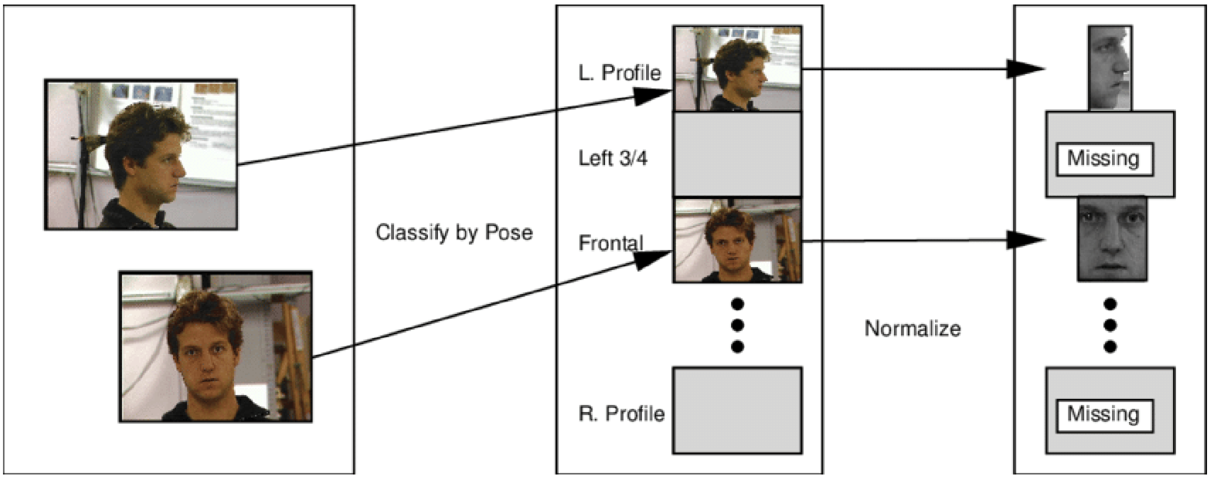
Model Evaluation:

After training, the model is evaluated on the test set. This is important to ensure the model’s ability to generalize to unseen images.

You measure the model’s accuracy, how often it correctly classifies new images as "Cat" or "Dog," Prediction:

VECTORIZATION

Vectorization is the process of converting a set of images of a face into a light-field vector. Vectorization is performed by first classifying each input image into one of a finite number of poses. For each pose, a normalization is then applied to convert the image into a sub-vector of the light-field vector.



CHAPTER-5

SYSTEM ANALYSIS

5.1 EXISTING SYSTEM

Facial recognition technology relies on the intricate patterns of a person's facial characteristics, such as the arrangement of eyes, nose, mouth, and other distinguishing features. These systems capture and process facial data using sophisticated algorithms, enabling them to perform tasks like identifying individuals in a crowd, verifying identities for secure access, or even assisting in law enforcement efforts. The possibilities presented by facial recognition are substantial, but they also come with significant responsibilities and considerations. Striking a balance between the convenience and utility of facial recognition and safeguarding individual privacy and civil liberties is an ongoing challenge. Ethical and legal issues surrounding data privacy, bias mitigation, and transparency have sparked intense debates and have led to the development of guidelines and regulations to govern the use of this technology.

DISADVANTANTAGES OF EXISTING SYSTEM:

Facial recognition technology can be used to invade privacy and for mass surveillance.

It can be use to track peoples movements , behaviour, and activities in real time without Their knowledge or consent.

Facial recognition system can be biased and misidentify members of minority groups.

Facial recognition technology stores and accesses personal biometric data, which can be misused or accessed by unauthorized people.

Facial recognition can be misused for fraud and other crimes. Third party companies can own pictures of peoples faces and use them to fuel their database.

5.2 PROPOSED SYSYTEM:

Face tracking is a part of face recognition system. Here we can use some systems algorithms to pick out specific, distinctive details about a human face. Facial recognition systems designed to detect and interpret facial expressions serve a multifaceted array of purposes across diverse domains. These systems primarily excel in understanding and decoding human emotions, offering applications ranging from healthcare to security, education, retail, and entertainment. By accurately identifying emotions like happiness, sadness, anger, and surprise, these systems facilitate enhanced human computer interaction, aid in healthcare diagnostics, improve security measures by detecting distress or suspicious behaviour, gauge student engagement in educational settings, provide valuable insights for retail and marketing, and enhance gaming experiences. Additionally, they hold potential in law enforcement and forensics by aiding in suspect identification or assessing veracity through micro expression analysis. emotional data.

ADVANTAGES OF PROPOSED SYSTEM:

➢ Bank(Incorrect Pin)

➢ Ration Shop (Eye Recognition)

➢ Airport (Used at airport to verify traveller’s identity)

ALGORITHMS USED:

➢ KLT(Kanade-lucas-Tomasi Algorithm is a Feature Tracking algorithm That is Used In computer Vision)

➢ PCA(principal component analysis is a technique used to simplify complex data by reducing the number of variables. PCA helps identify the most important features of an image ,like the l\key shapes and patterns that distinguish one face from another .

➢ VIOLA- JONES(used to recognize and detect objects, including this human face).

CHAPTER-6

MODULES

IMPLEMENTATION

6.1 MODULES

- Image acquisition
- Preprocessing
- Face detection
- Feature extraction
- Face representation
- Face matching or classification
- Decision making

6.2 MODULE DESCRIPTION

IMAGE ACQUISITION

Image acquisition is the process of capturing visual data, typically in the form of images or video frames, using devices like cameras or sensors. It serves as the initial step in various image processing systems, including face recognition. The captured images provide raw visual information that is later processed to extract meaningful features. The quality of the image, including factors like resolution, lighting, and camera angle, plays a significant role in the effectiveness of the subsequent steps in face recognition. Image acquisition can occur in real-time, where continuous frames are captured and analyzed for immediate processing, or in a more controlled manner, depending on the application.

Preprocessing

Preprocessing in a face recognition system involves preparing and enhancing the captured image to ensure that it is suitable for the subsequent steps of face detection and recognition. This step typically includes tasks like face detection, where the system locates the face within the image, and face alignment, where the face is adjusted to a consistent orientation. Image normalization may also be performed to adjust the lighting, contrast, and brightness, ensuring the face is clearly visible despite variations in environmental conditions. Additionally, facial feature enhancement can be applied to improve the clarity of important facial features like eyes, nose, and mouth. Overall, preprocessing helps to reduce noise, correct distortions, and standardize the input image to improve the accuracy of the face recognition process.

Face detection

Face detection is the process of identifying and locating faces within an image or video. The goal is to isolate the regions of interest that contain faces, allowing the system to focus on those areas for further processing. This step typically involves analysing the image for specific patterns, such as the arrangement of facial features (eyes, nose, mouth, etc.) or using machine learning algorithms that have been trained to recognize human faces. Various techniques, such as Haar cascades, Histogram of Oriented Gradients (HOG), or deep learning-based methods like convolutional neural networks (CNNs), can be employed for this task. Once the face or faces are detected, the system can then extract relevant features for recognition or verification. The accuracy of face detection is critical for the overall performance of the system, as it ensures that only the relevant parts of the image are processed.

Face matching or classification

Face matching or classification is the process of comparing the extracted facial features from a detected face with those stored in a database to identify or verify the individual. In this step, the system takes the numerical representation or feature vector derived from the face and compares it against pre-existing feature vectors of known faces. The comparison may be done using various algorithms that measure the similarity between the feature vectors, such as Euclidean distance, cosine similarity, or more advanced machine learning models. In a classification setup, the system assigns the detected face to a specific identity class based on the closest match, whereas in a verification setup, it checks whether the detected face matches a particular individual's stored data.

Face extraction

Face extraction is the process of identifying and isolating the unique features of a face from the captured image to create a distinctive representation of that face. This module focuses on extracting key facial landmarks such as the eyes, nose, mouth, and the overall facial structure. Various techniques are used to perform this task, including traditional methods like Local Binary Patterns (LBP) and Principal Component Analysis (PCA), as well as deep learning approaches such as Convolutional Neural Networks (CNNs), which automatically learn to identify and extract the most relevant features for face recognition. The extracted features are then transformed into a numerical or vector representation, often referred to as a face embedding, that captures the unique characteristics of the face. This representation is used in later stages of the face recognition system for matching or classification. The accuracy and robustness of the face extraction module are crucial, as it directly impacts the system's ability to differentiate between different individuals.

Decision making

The decision-making module in a face recognition system evaluates the results from the face matching or classification process and makes a final determination about the identity of the person. This module typically compares the similarity score or distance between the detected face's features and those stored in the database. If the score or match meets a predefined threshold, the system will either confirm the identity (in the case of verification) or identify the person (in the case of identification). In some systems, this module may also involve additional steps such as scoring the confidence level of the match, applying filtering rules, or flagging uncertain or ambiguous results for further review. The effectiveness of the decision-making process relies on the accuracy of the previous stages, as well as the threshold setting, to balance between false positives (incorrect matches) and false negatives (missed matches).

CHAPTER-7

SYSTEM DESIGN

7.1 FACE RECOGNITION ARCHITECTURE

Face recognition architecture involves several stages: face detection isolates the face, followed by alignment for standardization. Preprocessing, like normalization, enhances model training. Feature extraction using CNNs creates unique embeddings of the face. These embeddings are compared using distance metrics, and machine learning models like SVM or KNN classify them. Post-processing applies thresholds to determine if the face matches a known individual or is unknown. This process is used in applications such as security and authentication.

FIGURE 7.1.1 FACE RECOGNITION ARCHITECTURE

Face identification: face detection is the task of matching a given face image to one in an existing database of faces. It is the second part of face recognition (the first part being detection). It is a one-to-many mapping: you have to find an unknown person in a database to find who that person.

Face verification: face verification is a machine learning task in computer vision that involves determining whether two facial images belong to the same person or not. The task involves extracting features from the facial images, such as the shape and texture of the face, and then using these features to compare and verify the similarity between the images.

Comparision and identification: is a process that involves analyzing two or more items to determine if they match or are from the same source. It can be used in a variety of fields, including forensics, law enforcement, and industrial processes

Feature extraction: A feature is an individual measurable property within a recorded dataset. In machine learning and statistics, features are often called “variables” or “attributes.” Relevant features have a correlation or bearing on a model’s use case. In a patient medical dataset, features could be age, gender, blood pressure, cholesterol level, and other observed characteristics relevant to the patient.

Face detection and cropping : is a process that uses computer vision to identify faces in an image and then crop them out. It can be used to create clean headshots, ID cards, and more.

Test Image: Used for evaluating or testing how well a model or system works, typically by processing an image it hasn't encountered before.

Input Query: The data or question that is provided to a system, which can include images, text, or other forms of input, that the system processes to generate an output.

7.2 PHASE IN PATTERN RECOGNITION

Pattern recognition typically involves several phases. First, data acquisition captures the raw data, followed by preprocessing to clean and normalize it. Then, feature extraction identifies relevant patterns or characteristics from the data. Afterward, classification or decision-making algorithms are applied to label or categorize the data. Finally, post-processing refines the results, ensuring accuracy and making adjustments if necessary.

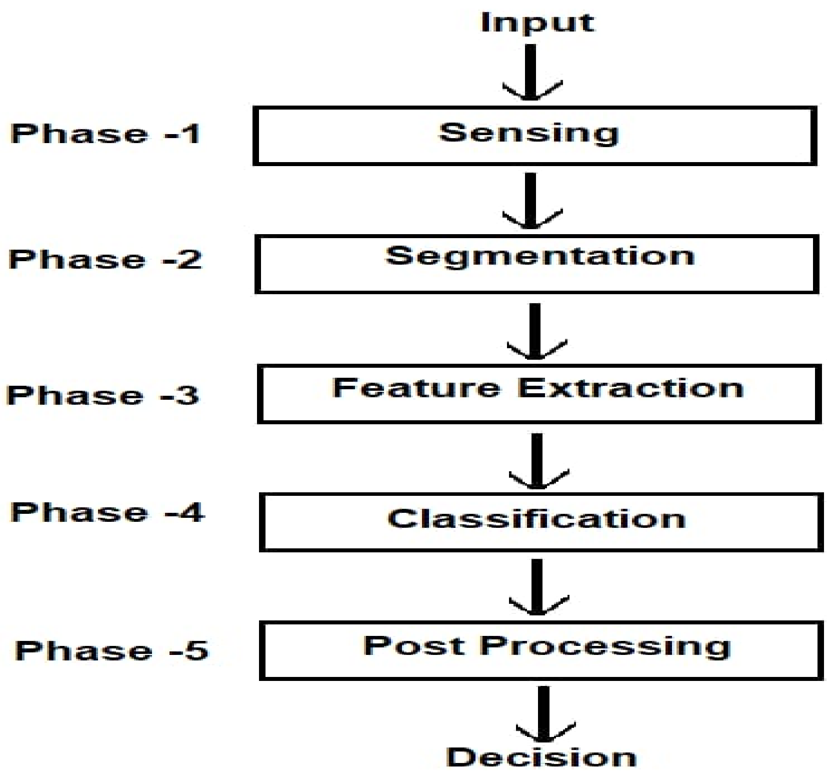


FIGURE 7.2.1 PHASE IN PATTERN RECOGNITION

Sensing: It deals with problem arises in the input such as its bandwidth, resolution, sensitivity, distortion, signal-to-noise ratio, latency, etc

Segmentation and Grouping: Deepest problems in pattern recognition that deals with the problem of recognizing or grouping together the various parts of an object.

Feature Extraction: It deals with the characterization of an object so that it can be recognized easily by measurements. Those objects whose values are very similar for the objects are considered to be in the same category, while those whose values are quite different for the objects are placed in different categories.

Classification: It deals with assigning the object to their particular categories by using the feature vector provided by the feature extractor and determining the values of all of the features for a particular input.

Post Processing: It deals with action decision-making by using the output of the classifier. Action such as minimum-error-rate classification will minimize the total expected cost.

7.3 MAIN STEPS IN PATTERN RECOGNITION

Pattern recognition involves collecting raw data, followed by preprocessing to clean and standardize it. Relevant features are then extracted from the data to capture important patterns. Classification algorithms are applied to label the features into categories, and post-processing is done to refine the results and ensure accuracy.

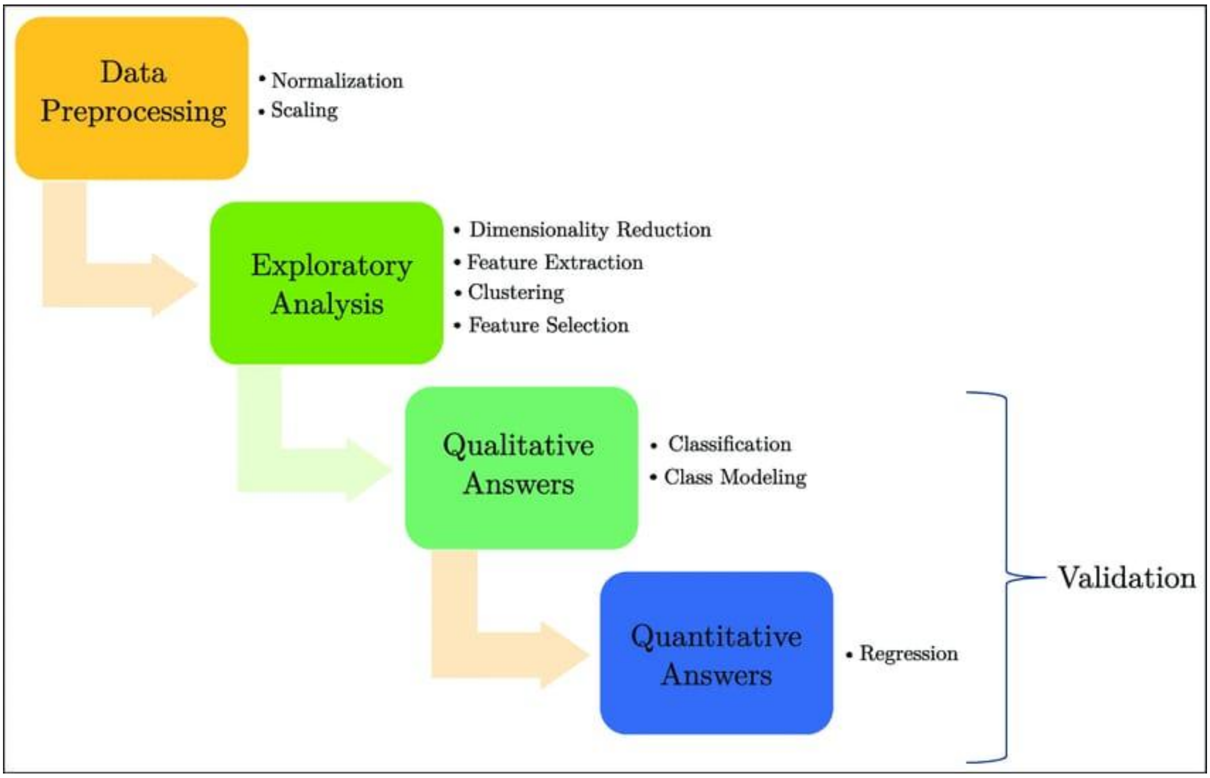


FIGURE 7.3.1 MAIN STEPS PATTERN RECOGNITION

7.4 DATAFLOW DIAGRAMS

A Data Flow Diagram (DFD) of a Face Recognition System is a visual representation that illustrates how data moves through the system, showing the various processes, inputs, outputs, and interactions within the system. It highlights how different components, such as face detection, feature extraction, and face matching, process the data as it flows through the system. The DFD identifies external entities like users, cameras, and databases, and it shows how data is exchanged between them. The diagram helps in understanding the flow of information from capturing a face image to processing it and returning the recognition results, providing a clear map of the system's operations.

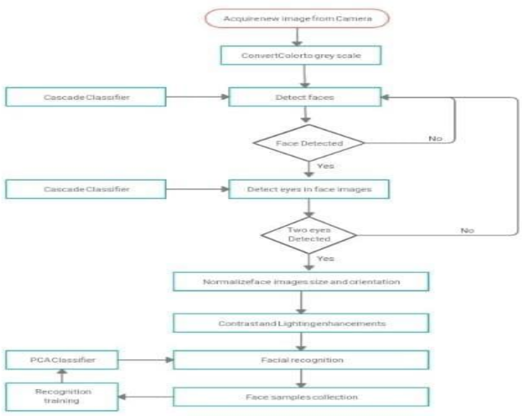


FIGURE 7.4.1 DATA FLOW DIAGRAMS

CHAPTER-8

SOURCE CODE

FACE RECOGINITION HOME PAGE

```html
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Face Detection</title>
</head>
<body>
   <h1>Face Detection Application</h1>
   <form action="/upload" method="post" enctype="multipart/form-data">
      <label for="file">Upload an Image:</label>
      <input type="file" name="file" id="file" accept="image/*">
      <button type="submit">Upload</button>
   </form>
</body>
</html>
```

FACE DECTECION

```
import cv2

# Load the pre-trained face detection model (Haar Cascade Classifier)

face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +
'haarcascade_frontalface_default.xml')


# Read an image from file

image = cv2.imread("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\geetu.jpg")  # Change
the path to your image file


# Convert the image to grayscale (required for face detection)

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


# Detect faces in the image

faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30,
30))


# Draw rectangles around the faces

for (x, y, w, h) in faces:

    cv2.rectangle(image, (x, y), (x + w, y + h), (0, 255, 0), 2)


# Display the image with detected faces

cv2.imshow('Face Detection', image)


# Wait for a key press to close the window

cv2.waitKey(0)

cv2.destroyAllWindows()
```

EDGE DECTECTION

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image from disk.
img = cv2.imread("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\Ganeshji.webp")
# Convert BGR image to RGB
image_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Apply Canny edge detection
edges = cv2.Canny(image= image_rgb, threshold1=100, threshold2=700)

# Create subplots
fig, axs = plt.subplots(1, 2, figsize=(7, 4))

# Plot the original image
axs[0].imshow(image_rgb)
axs[0].set_title('Original Image')

# Plot the blurred image
axs[1].imshow(edges)
axs[1].set_title('Image edges')

# Remove ticks from the subplots
for ax in axs:
```

```
    ax.set_xticks([])
    ax.set_yticks([])

# Display the subplots
plt.tight_layout()
plt.show()
```

SEGMENTATION

```
import cv2
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Load image
image = cv2.imread("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\cat.jpg")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# 1. Thresholding Segmentation
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
_, thresh_img = cv2.threshold(gray, 120, 255, cv2.THRESH_BINARY)

# 2. Edge Detection using Canny
edges = cv2.Canny(gray, 100, 200)

# 3. K-Means Clustering for Color-based Segmentation
reshaped_image = image_rgb.reshape((-1, 3))
kmeans = KMeans(n_clusters=3)
kmeans.fit(reshaped_image)
segmented_image = kmeans.cluster_centers_[kmeans.labels_]
segmented_image = segmented_image.reshape(image_rgb.shape).astype(np.uint8)
```

```
# Plot results
fig, axs = plt.subplots(2, 2, figsize=(10, 10))

# Original Image
axs[0, 0].imshow(image_rgb)
axs[0, 0].set_title('Original Image')
axs[0, 0].axis('off')

# Thresholding Result
axs[0, 1].imshow(thresh_img, cmap='gray')
axs[0, 1].set_title('Thresholding Segmentation')
axs[0, 1].axis('off')

# Edge Detection Result
axs[1, 0].imshow(edges, cmap='gray')
axs[1, 0].set_title('Canny Edge Detection')
axs[1, 0].axis('off')

# K-Means Segmentation Result
axs[1, 1].imshow(segmented_image)
axs[1, 1].set_title('K-Means Segmentation')
axs[1, 1].axis('off')

plt.show()
```

ENHANCEMENT

```
import cv2
import numpy as np
from PIL import Image, ImageEnhance

# Load an image using OpenCV
image = cv2.imread("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\dog.jpg")   # Replace with your image path

# Convert BGR image (OpenCV format) to RGB (PIL format) for use with Pillow
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
pil_image = Image.fromarray(image_rgb)

# 1. *Histogram Equalization* (Improve contrast)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
equalized_image = cv2.equalizeHist(gray)

# 2. *Gamma Correction* (Adjust brightness)
gamma = 1.5  # Higher values darken the image; values less than 1 brighten it
gamma_corrected_image = np.array(255 * (image / 255) ** gamma, dtype='uint8')

# 3. *Image Sharpening* (Enhance edges)
kernel = np.array([[0, -1, 0],
                   [-1, 5,-1],
                   [0, -1, 0]])  # Sharpening kernel
sharpened_image = cv2.filter2D(image, -1, kernel
```

```
# 4. *Brightness and Contrast Adjustment* (Using PIL)
enhancer = ImageEnhance.Brightness(pil_image)
brightened_image = enhancer.enhance(1.5)  # 1.0 is original brightness, >1.0 increases
brightness

contrast_enhancer = ImageEnhance.Contrast(brightened_image)
contrast_image = contrast_enhancer.enhance(1.5)  # Increase contrast

# 5. *Smoothing (Blurring)* (Reduce noise or smooth the image)
blurred_image = cv2.GaussianBlur(image, (5, 5), 0)  # Gaussian blur with kernel size (5,5)

# 6. *Edge Detection* (Using Canny edge detector)
edges = cv2.Canny(image, 100, 200)

# Display the images using OpenCV
cv2.imshow('Original Image', image)
cv2.imshow('Histogram Equalization', equalized_image)
cv2.imshow('Gamma Correction', gamma_corrected_image)
cv2.imshow('Sharpened Image', sharpened_image)
cv2.imshow('Brightness and Contrast Adjustment', np.array(contrast_image))
cv2.imshow('Blurred Image', blurred_image)
cv2.imshow('Edge Detection', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
dsize=(new_width_1, new_height_1),
                    interpolation=cv2.INTER_AREA)
```

IMAGE ROTATION

```
import cv2
import matplotlib.pyplot as plt

# Read image from disk.
img = cv2.imread("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\Ganeshji.webp")

# Convert BGR image to RGB
image_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Image rotation parameter
center = (image_rgb.shape[1] // 2, image_rgb.shape[0] // 2)
angle = 30
scale = 1

# getRotationMatrix2D creates a matrix needed for transformation.
rotation_matrix = cv2.getRotationMatrix2D(center, angle, scale)

# We want matrix for rotation w.r.t center to 30 degree without scaling.
rotated_image = cv2.warpAffine(image_rgb, rotation_matrix, (img.shape[1], img.shape[0]))
```

```
# Create subplots
fig, axs = plt.subplots(1, 2, figsize=(7, 4))
# Plot the original image
axs[0].imshow(image_rgb)
axs[0].set_title('Original Image')

# Plot the Rotated image
axs[1].imshow(rotated_image)
axs[1].set_title('Image Rotation')

# Remove ticks from the subplots
for ax in axs:
    ax.set_xticks([])
    ax.set_yticks([])

# Display the subplots
plt.tight_layout()
plt.show()
```

IMAGE COMPRESSION

```
import cv2

# Load the image

image = cv2.imread("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\dog.jpg")    # Replace with your image path


# Compress the image using JPEG (lossy compression)

compression_quality = 50  # Range from 0 (low quality) to 100 (high quality)

encode_param = [int(cv2.IMWRITE_JPEG_QUALITY), compression_quality]


# Compress and save the image

cv2.imwrite('compressed_image.jpg', image, encode_param)


# Print the file size before and after compression

import os

original_size = os.path.getsize("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\dog.jpg")  # Original size in bytes

compressed_size = os.path.getsize("C:\\Users\\pooja\\OneDrive\\Desktop\\Pictures\\dog.jpg") # Compressed size in bytes


print(f"Original size:  {original_size / 1024:.2f}  KB")

print(f"Compressed size:  {compressed_size / 1024:.2f}  KB")
```

CHAPTER – 9

SYSTEM STUDY

9.1 FEASIBILTY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are,

♦ **ECONOMICAL FEASIBILITY**

♦ **TECHNICAL FEASIBILITY**

♦ **SOCIAL FEASIBILITY**

9.1.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

9.1.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

9.1.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER-10

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

10.1 SYSTEM TESTING

10.1.1Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

10.1.2 Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

10.1.3Functional testing

Functional testing provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

10.2 System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

10.2.1White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

10.2.2Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

2. Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed

3.Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects. The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

4. Acceptance Testing

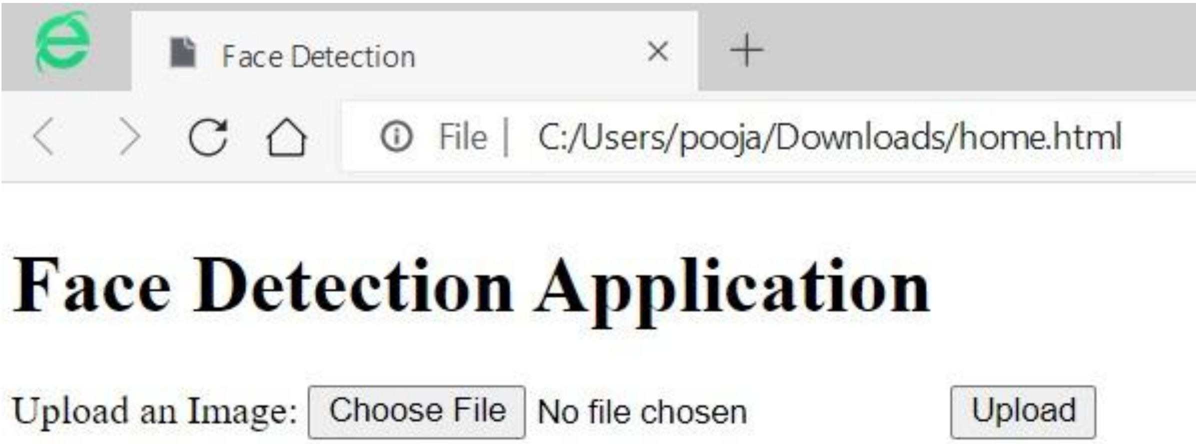
User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

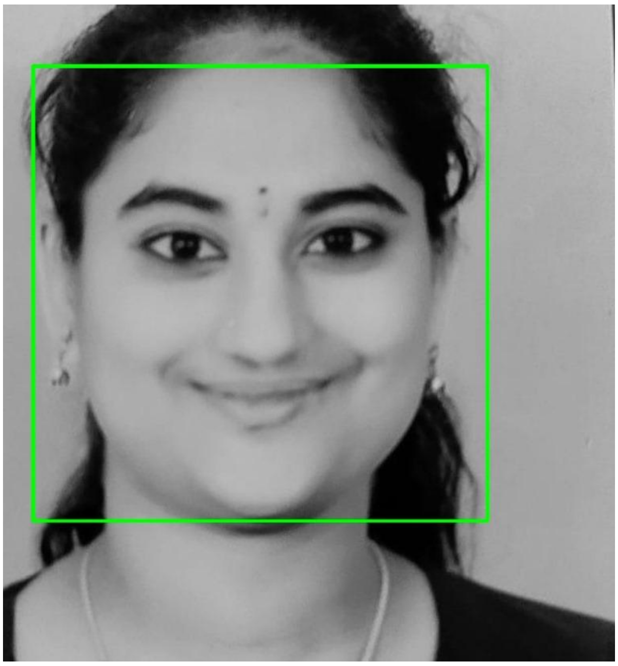
S.no	Test Case	Excepted Result	Result	Remarks(IF Fails)
1	User Register	If User registration successfully.	Pass	If already user email exist then it fails.
2	User Login	If Username and password is correct then it will getting valid page.	Pass	Un Register Users will not logged in.
3	Random forest and svm	The request will be accepted by the random forest and svm	Pass	The request will be accepted by the random forest and svm otherwise its failed
4	Decision Tree and multilayer perceptron	The request will be accepted by the Decision Tree and multilayer perceptron	Pass	The request will be accepted by the Decision Tree and multilayer perceptron otherwise its failed
5	Naive Bayes and k-nearest neighbour	The request will be accepted by the Naive Bayes and k-nearest neighbour	Pass	The request will be accepted by the Naive Bayes and k-nearest neighbour otherwise its failed
6	View dataset by user	Data set will be displayed by the user	Pass	Results not true failed
7	User classification	Display reviews with true results	Pass	Results not true failed
8	Calculate accuracy macro avg and weighted avg	macro avg and weighted avg calculated	Pass	macro avg and weighted avg not displayed failed
9	Admin login	Admin can login with his login credential. If success he get his home page	Pass	Invalid login details will not allowed here
10	Admin can activate the register users	Admin can activate the register user id	Pass	If user id not found then it won’t login.

CHAPTER-11

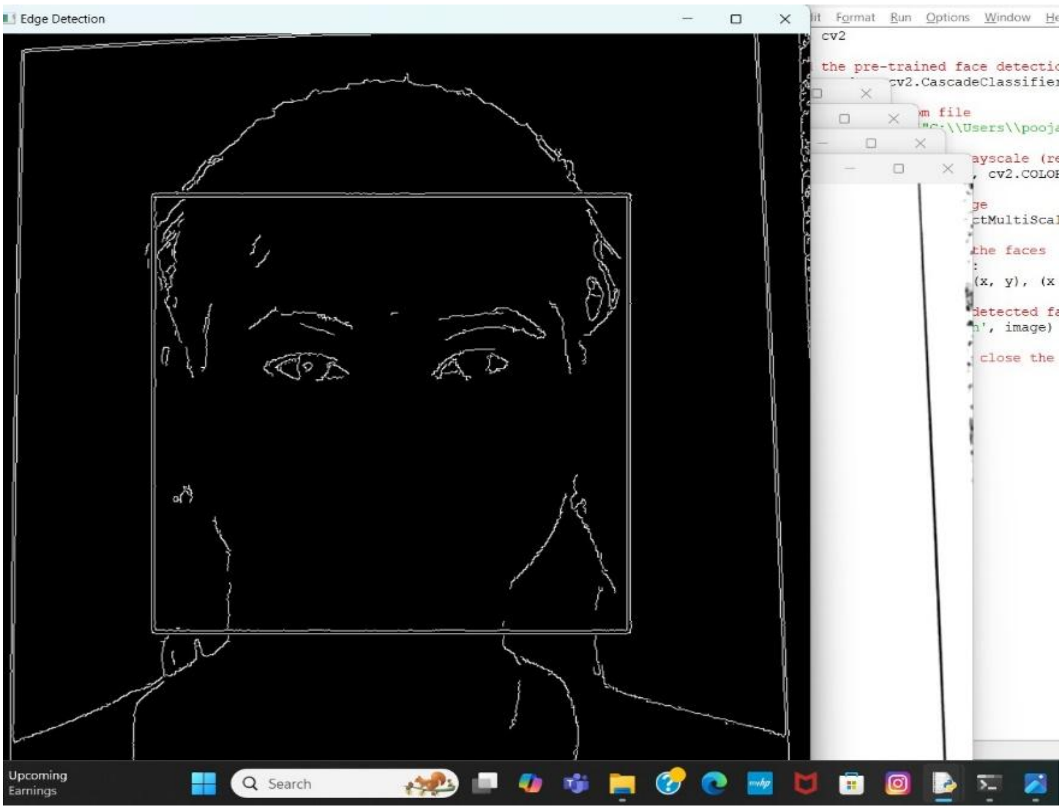
OUTPUT SCREENS



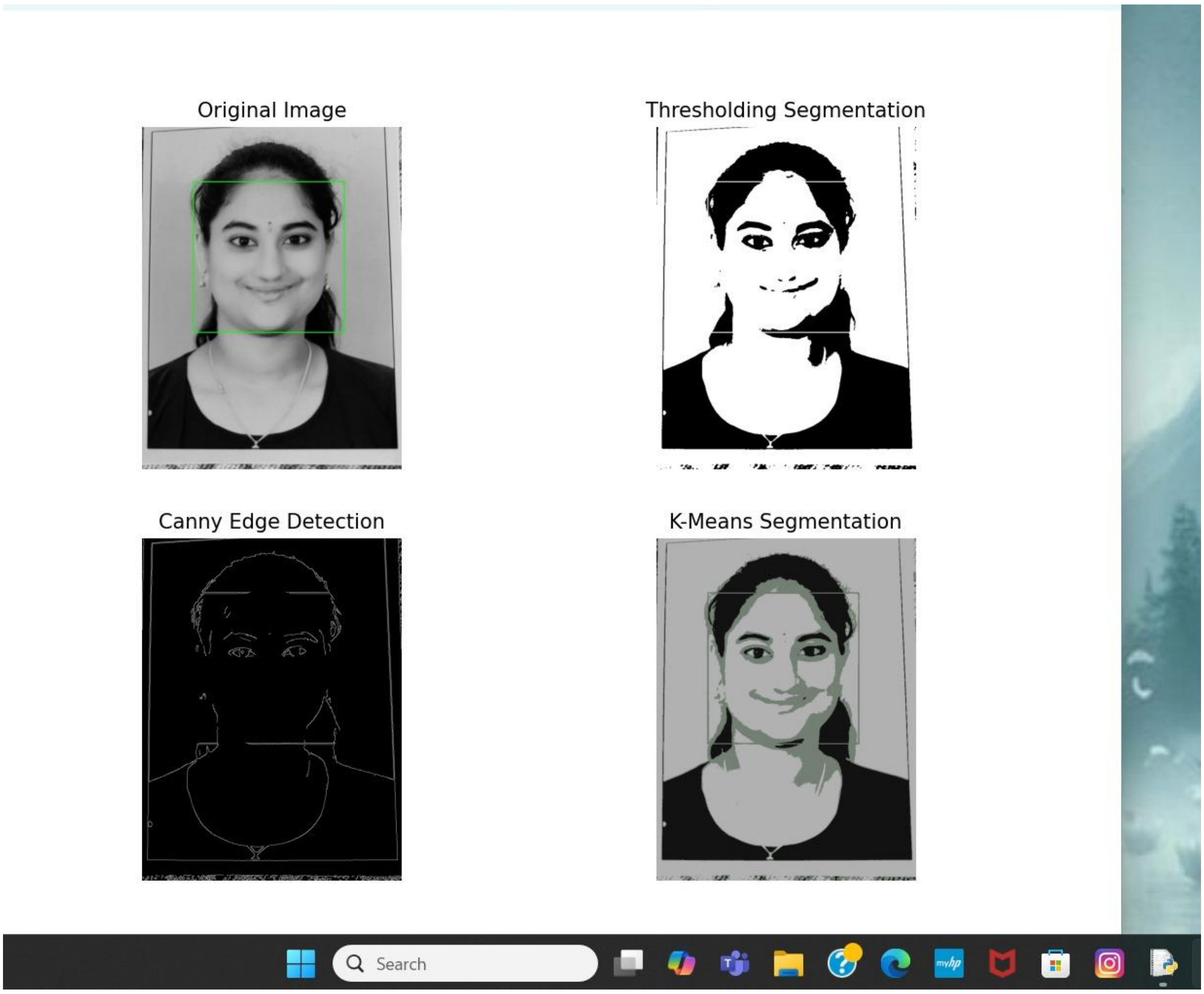
OUTPUT SCREEN 11.1 HOME PAGE



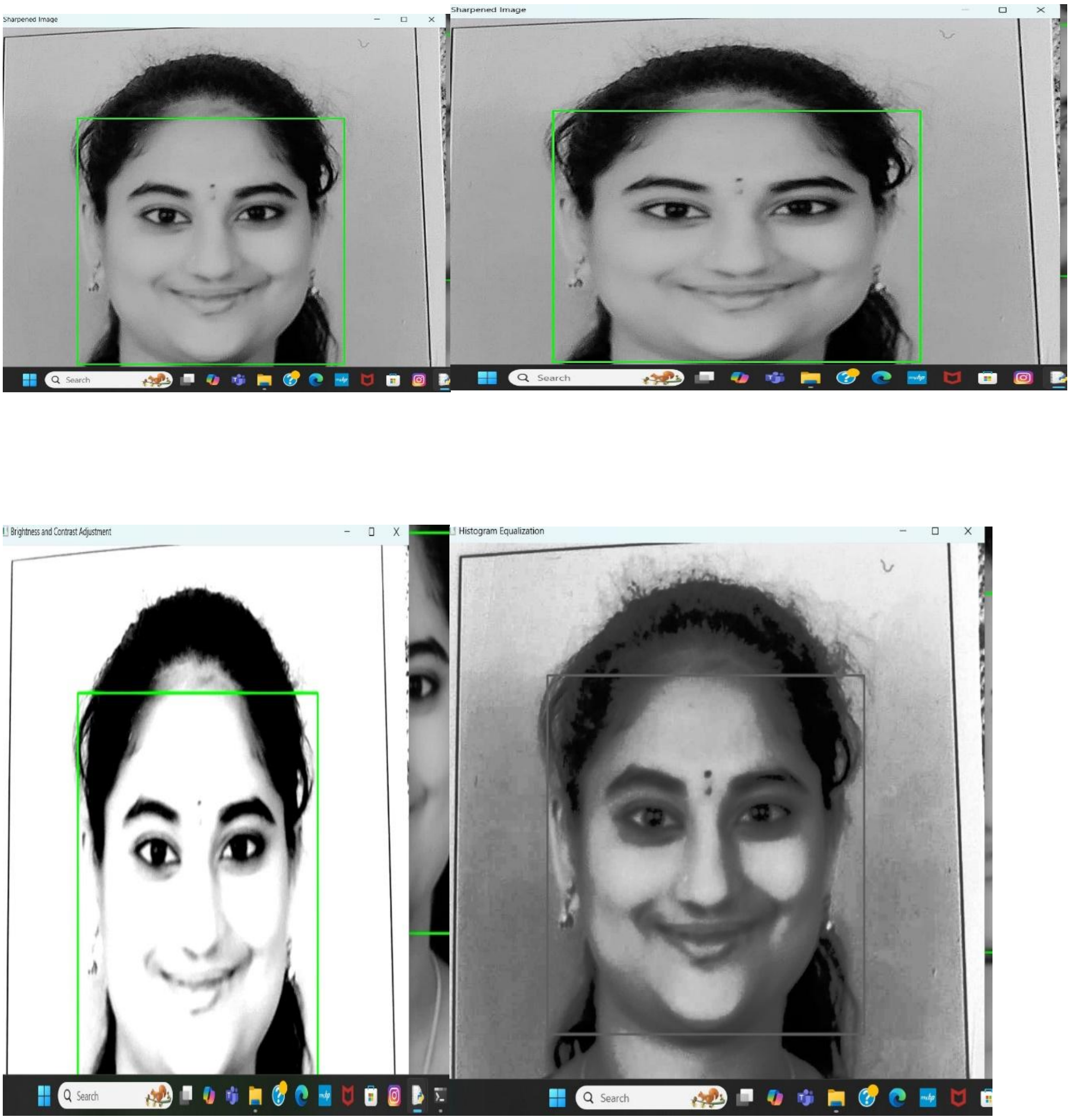
OUTPUT SCREEN 11.2: FACE DETECTION



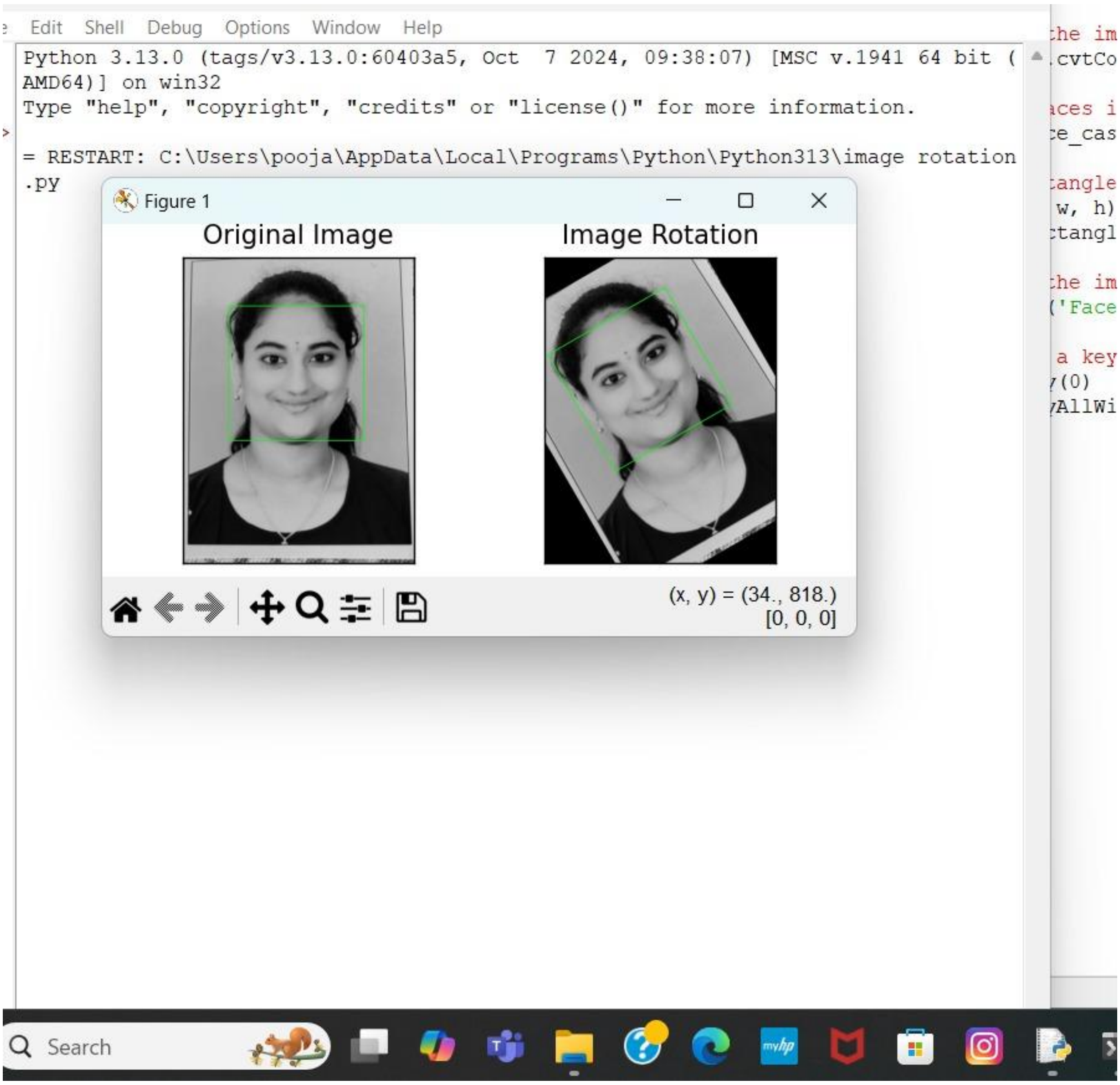
OUTPUT SCREEN 11.3: EDGE DETECTION



OUTPUT SCREEN 11.4: SEGMENTATION



OUTPUT SCREEN 11.5 ENHANCEMENT



OUTPUT SCREEN 11.6 ROTATION

```
========

= RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image enhancement.py =

== RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image processing.py =
Traceback (most recent call last):
  File "C:\Users\pooja\AppData\Local\Programs\Python\Python313\image processing.py", lin
e 26, in <module>
    dsize =(new_width500, new_height250),
NameError: name 'new_width500' is not defined. Did you mean: 'new_width'?

===== RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image pre.py =====

=== RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image rotation.py ==

= RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image transformation.p
y

= RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image transformation.p
y

= RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image transformation.p
y

= RESTART: C:\Users\pooja\AppData\Local\Programs\Python\Python313\image compression.py =
Original size: 57.71 KB
Compressed size: 57.71 KB
```

Ln: 100 Col: 0

OUTPUT SCREEN 11.7 IMAGE COMPRESSION

CHAPTER-12

CONCLUSION

Face recognition technology has revolutionized various industries, offering improved security, convenience, and efficiency. With its numerous applications, face recognition has become an essential tool in law enforcement, access control, identity verification, and more. While challenges and concerns surrounding bias, privacy, and security remain, ongoing research and advancements in algorithms and integration with emerging technologies will continue to enhance the accuracy, reliability, and acceptability of face recognition systems. Ultimately, face recognition technology has the potential to transform the way we interact with devices, access information, and ensure public safety, making it an exciting and rapidly evolving field.

CHAPTER-13

FUTURE ENHANCEMENT

Future Scope

•**Government/ Identity Management:** Governments all around the world are using face recognition systems to identify civilians. America has one of the largest face databases in the world, containing data of about 117 million people.

•**Emotion & Sentiment Analysis:** Face Detection and Recognition have brought us closer to the technology of automated psyche evaluation. As systems now a days can judge the precise emotions frame by frame in order to evaluate the psyche.

•**Authentication systems:** Various devices like mobile phones or even ATMs work using facial recognition, thus making getting access or verification quicker and hassle free.

•**Full Automation:** This technology helps us become fully automated as there is very little to zero amount of effort required for verification using facial recognition.

•**High Accuracy:** Face Detection and Recognition systems these days have developed very high accuracy and can be trained using very small data sets and the false acceptance rates have dropped down significantly.

5.2 Limitations

•**Data Storage:** Extensive data storage is required for creating, training and maintaining big face databases which is not always feasible.

•**Computational Power:** The requirement of computational power also increases with increase in the size of the database. This becomes financially out of bounds for smaller organizations.These conditions may not always be therefore
Creating a major drawback.

CHAPTER-14

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